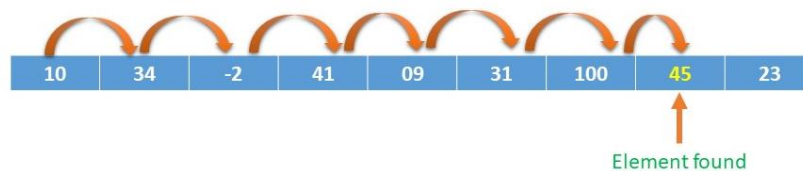




Linear Search

The following algorithm takes inputs an Array A of size N and key element. Further, it searches for the key element in the given list [Array A].

Key element=45



In the above diagram, key element is 45 to be searched in the given list of array elements. The algorithm scans each element sequentially from an array and compares with key element to be searched. It terminates the element match found or else it continues to search until the last element of the given array.

Algorithm:

//**key** is the element to be searched in the given list of elements.

// **A[N]** is an array of size N.

Linear_Search(A,N)

{

 For I:=1 to N step 1 Do

 IF key = A[I] then

 Display "Element found "

 Exit.

 [END IF]

Done

}

Binary Search

The searching algorithm goes on dividing the sorted list of elements by half and searching in sub lists based on the mid value found. The following diagram Illustrates about the binary search.



	0	1	2	3	4	5	6	7	8	9
Search 23	2	5	8	12	16	23	38	56	72	91
	L=0	1	2	3	M=4	5	6	7	8	H=9
23 > 16 take 2 nd half	2	5	8	12	16	23	38	56	72	91
	0	1	2	3	4	L=5	6	M=7	8	H=9
23 < 56 take 1 st half	2	5	8	12	16	23	38	56	72	91
	0	1	2	3	4	L=5, M=5	H=6	7	8	9
Found 23, Return 5	2	5	8	12	16	23	38	56	72	91

Algorithm:

//**key** is the element to be searched in the given list of elements. and **A[N]** is an array size N.

//**low and high** are the references to the beginning and ending of the list Initially.

// **mid** is the reference to the middle of the list.

Binary_Search(A,key,)

{

Set low and high to the beginning and ending indexes of array A.

WHILE low ≤ high Do

BEGIN

$$mid = \frac{low+high}{2}$$

IF key = A[mid] THEN

print 'Element found'

exit

ELSE IF key < A[mid] THEN

high=mid-1

ELSE

low=mid+1

[END IF]

END LOOP }